

Linux Firewall and Secure Remote Access

Laboratory 4

Student Practical Laboratory Manual

Course	CORE102 TCP IP Networking and Linux Network Tools
Instructor	Aminu Idris AMCPN
Maximum score	100 marks
Gradebook category	Practical Laboratories
Estimated duration	5 to 6 hours
Platform	Kali Linux or Ubuntu Linux in an authorised laboratory environment
Version	September 2026

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Purpose and Scenario

This laboratory develops the defensive skills required to identify exposed services, enforce host firewall rules and configure SSH key authentication. Every change includes a validation and recovery step so that security controls remain observable and reversible.

You are hardening a Linux server before it joins an internal training network. The system currently exposes services without a documented policy. You must establish a baseline, permit only approved access, test the controls from a client and secure administrative access with an SSH key.

Learning Outcomes

- Identify listening TCP and UDP services and the processes that own them.
- Explain INPUT, OUTPUT and FORWARD traffic in Linux packet filtering.
- Create, verify, order and remove UFW rules safely.
- Inspect iptables or nftables compatibility rules and counters.
- Generate and deploy an Ed25519 SSH key pair.
- Validate SSH configuration before reloading the service.
- Test firewall and SSH controls from a separate authorised client.
- Document rollback steps and residual risk.

Environment and Prerequisites

- A server VM and a client VM on 192.168.56.0/24 are recommended.
- Console access to the server VM and a fresh snapshot.
- sudo permission and OpenSSH server installed on the server.
- UFW, iptables, netcat and Python 3.
- A confirmed client-to-server route before firewall changes.

Safety and Authorisation

- Do not perform firewall changes on a remote-only system without console recovery access.
- Allow and test SSH before enabling a default-deny firewall policy.
- Keep one authenticated SSH session open while testing a new SSH configuration in a second session.
- Never submit a private SSH key. Only the public key fingerprint may appear in evidence.
- Apply rules only to the authorised laboratory hosts and remove temporary services during cleanup.

Evidence Requirements

Item	Required evidence	Included
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1	Baseline service and socket inventory.	[]
2	UFW policy and numbered-rule evidence before and after testing.	[]
3	Client-side allowed and denied connection tests.	[]
4	Safe iptables custom-chain exercise and cleanup.	[]
5	SSH public-key setup, fingerprint, successful login and server validation.	[]
6	Final exposure review and rollback record.	[]

Laboratory Topology

System	Example address	Purpose
Server VM	192.168.56.10	UFW, service inventory and SSH hardening
Client VM	192.168.56.20	Authorised connectivity and rule testing
Private network	192.168.56.0/24	Host-only or internal virtual network

Part 1 Baseline and Recovery Preparation

Port and protocol	Bind address	Process	Expected purpose	Keep or remove
22/tcp	0.0.0.0	sshd	secure remove administration (SSH)	keep

22/tcp	[::]	sshd	Secure remote administration (SSH IPv6)	keep
53/tcp+53/udp	127.0.0.53/127.0.0.54	systemd- resolve	Local DNS resolver	keep(local only)
68/udp	192.168.56.101	systemd- networkd	DHCP client	keep
323/udp	127.0.0.1/[::1]	chronyd	NTP time synchronization	keep(local only)

```
File Machine View Input Devices Help
kali@kali: ~/core102_lab4
Session Actions Edit View Help
kali@kali:~/core102_lab4$ mkdir -p ~/core102_lab4 06 cd ~/core102_lab4
kali@kali:~/core102_lab4$ date
Tue Sep 29 09:35:11 AM EDT 2026
kali@kali:~/core102_lab4$ hostname
kali
kali@kali:~/core102_lab4$ ip -br addr
lo UNKNOWN 127.0.0.1/8 ::1/128
eth0 UP 192.168.56.102/24 fe80::a00:27ff:fe5f:cc31/64
kali@kali:~/core102_lab4$ ip route
192.168.56.0/24 dev eth0 proto kernel scope link src 192.168.56.102 metric 100
kali@kali:~/core102_lab4$ sudo ss -tulpn
[sudo] password for kali:
State      Recv-Q    Send-Q   Local Address:Port   Peer Address:Port   Process
tcp        LISTEN     0         128          0.0.0.0:22          0.0.0.0:*             users:((("sshd",pid=585,fd=6))
tcp        LISTEN     0         128          :::::22             :::::*                 users:((("sshd",pid=585,fd=7))
kali@kali:~/core102_lab4$ sudo systemctl --state=running --no-pager
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
accounts-daemon.service             loaded active running Accounts Service
colord.service                      loaded active running Manage, Install and Generate Color Profiles
cron.service                       loaded active running Regular background program processing daemon
dbus.service                       loaded active running D-Bus System Message Bus
getty@tty1.service                 loaded active running Getty on tty1
haveged.service                    loaded active running Entropy Daemon based on the HAVEGE algorithm
lightdm.service                    loaded active running Light Display Manager
ModemManager.service               loaded active running Modem Manager
NetworkManager.service             loaded active running Network Manager
polkit.service                     loaded active running Authorization Manager
rtkit-daemon.service               loaded active running RealtimeKit Scheduling Policy Service
sshd.service                       loaded active running OpenSSH Secure Shell server
systemd-journald.service            loaded active running Journal Service
```

```
santi@ubutu: ~/  
Session Actions Edit View Help  
santi@ubutu:~$ mkdir -p ~/core102_lab4 && cd ~/core102_lab4  
santi@ubutu:~/core102_lab4$ date  
Tue Sep 29 02:06:42 PM UTC 2026  
santi@ubutu:~/core102_lab4$ hostname  
ubutu  
santi@ubutu:~/core102_lab4$ ip -br addr  
lo UNKNOWN 127.0.0.1/8 ::1/128  
enp0s3 UP 192.168.56.101/24 metric 1024 fe80::a00:27ff:fe7a:4925/64  
santi@ubutu:~/core102_lab4$ sudo ss -tulpn  
[sudo: authenticate] Password:  
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port  
Process  
udp UNCONN 0 0 127.0.0.54:53 0.0.0.0:*  
users:(("systemd-resolve",pid=586,fd=18))  
udp UNCONN 0 0 127.0.0.53%lo:53 0.0.0.0:*  
users:(("systemd-resolve",pid=586,fd=16))  
udp UNCONN 0 0 192.168.56.101%enp0s3:68 0.0.0.0:*  
users:(("systemd-network",pid=867,fd=36))  
udp UNCONN 0 0 127.0.0.1:323 0.0.0.0:*  
users:(("chronyd",pid=1145,fd=4))  
udp UNCONN 0 0 [::1]:323 [::]:*  
users:(("chronyd",pid=1145,fd=5))  
tcp LISTEN 0 4096 127.0.0.54:53 0.0.0.0:*  
users:(("systemd-resolve",pid=586,fd=19))  
tcp LISTEN 0 4096 0.0.0.0:22 0.0.0.0:*  
users:(("sshd",pid=1107,fd=3),("systemd",pid=1,fd=202))  
tcp LISTEN 0 4096 127.0.0.53%lo:53 0.0.0.0:*  
users:(("systemd-resolve",pid=586,fd=17))  
tcp LISTEN 0 4096 [::]:22 [::]:*  
users:(("sshd",pid=1107,fd=4),("systemd",pid=1,fd=203))  
santi@ubutu:~/core102_lab4$ sudo systemctl --type=service --state=running --no-pager  
UNIT LOAD ACTIVE SUB DESCRIPTION  
chrony.service loaded active running chrony, an NTP client/server  
cron.service loaded active running Regular background program processing ...  
dbus.service loaded active running D-Bus System Message Bus  
getty@tty1.service loaded active running Getty on tty1  
ModemManager.service loaded active running Modem Manager  
multipathd.service loaded active running Device-Mapper Multipath Device Control...  
networkd-dispatcher.service loaded active running Dispatcher daemon for systemd-networkd  
polkit.service loaded active running Authorization Manager  
rsyslog.service loaded active running System Logging Service  
snap.network-manager.networkmanager.se... loaded active running Service for snap application network-m...  
snapd.service loaded active running Snap Daemon
```

```
santi@ubuntu: ~/
Session Actions Edit View Help

ModemManager.service      loaded active running Modem Manager
multipathd.service        loaded active running Device-Mapper Multipath Device Control...
networkd-dispatcher.service loaded active running Dispatcher daemon for systemd-networkd
polkit.service            loaded active running Authorization Manager
rsyslog.service           loaded active running System Logging Service
snap.network-manager.networkmanager.se... loaded active running Service for snap application network-m...
snapd.service             loaded active running Snap Daemon
ssh.service               loaded active running OpenBSD Secure Shell server
systemd-journald.service   loaded active running Journal Service
systemd-logind.service     loaded active running User Login Management
systemd-networkd.service   loaded active running Network Management
systemd-resolved.service   loaded active running Network Name Resolution
systemd-udevd.service      loaded active running Rule-based Manager for Device Events a...
udisks2.service           loaded active running Disk Manager
unattended-upgrades.service loaded active running Unattended Upgrades Shutdown
user@1000.service          loaded active running User Manager for UID 1000

Legend: LOAD    → Reflects whether the unit definition was properly loaded.
ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
SUB    → The low-level unit activation state, values depend on unit type.

20 loaded units listed.
santi@ubuntu:~/core102_lab4$ sudo ufw status verbose
Status: inactive
santi@ubuntu:~/core102_lab4$ sudo iptables -L -n -v --line-numbers
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target     prot opt in     out     source destination
Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target     prot opt in     out     source destination
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target     prot opt in     out     source destination
santi@ubuntu:~/core102_lab4$ sudo ip6tables -L -n -v --line-numbers 2>/dev/null || true
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target     prot opt in     out     source destination
Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target     prot opt in     out     source destination
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target     prot opt in     out     source destination
santi@ubuntu:~/core102_lab4$
```

Report

Before making any firewall changes, I took a snapshot of the server VM (lab4-baseline-before-firewall) so I had a clean state to return to if a rule locked me out. I then recorded the server's interfaces, routes, listening sockets and running services.

The server, an Ubuntu VM at 192.168.56.101, had five listening entries. Only SSH on port 22 was open to the network, bound on both IPv4 (0.0.0.0) and IPv6 (:::). The rest were local or client-side. The DNS stub resolver (systemd-resolved, port 53) and the NTP service (chronyd, port 323) listen only on loopback addresses, so other machines can't reach them. The DHCP client on 68/udp is bound to the server's own interface address and only handles replies to requests the server sends out.

Since SSH is my only remote administration route and everything else is local, I decided to keep all five entries. No service needed to be stopped or disabled. That leaves port 22 as the only thing I need to protect, so the firewall plan is to allow SSH first and then apply a default-deny policy for incoming traffic, so I don't cut off my own access


```

santi@ubutu:~$ python3 -m http.server 8080 --bind 0.0.0.0
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
^C
Keyboard interrupt received, exiting.
santi@ubutu:~$ ls ~/core102_lab4/
08_ufw_before_test.txt  09a_ufw_rule_added.txt  09b_ufw_after_removal.txt  09c_ufw_after_restore.txt
santi@ubutu:~$ sudo iptables --version
sudo iptables -L -n -v --line-numbers | tee ~/core102_lab4/10_iptables_after_ufw.txt
iptables v1.8.11 (nf_tables)

Chain INPUT (policy DROP 6 packets, 360 bytes)
num  pkts bytes target     prot opt in     out     source            destination
1    4906 428K ufw-before-logging-input all  -- *      *        0.0.0.0/0        0.0.0.0/0
2    4906 428K ufw-before-input all  -- *      *        0.0.0.0/0        0.0.0.0/0
3      50 3772 ufw-after-input all  -- *      *        0.0.0.0/0        0.0.0.0/0
4       6 360 ufw-after-logging-input all  -- *      *        0.0.0.0/0        0.0.0.0/0
5       6 360 ufw-reject-input all  -- *      *        0.0.0.0/0        0.0.0.0/0
6       6 360 ufw-track-input all  -- *      *        0.0.0.0/0        0.0.0.0/0

Chain FORWARD (policy DROP 0 packets, 0 bytes)
num  pkts bytes target     prot opt in     out     source            destination
1     0 0 ufw-before-logging-forward all  -- *      *        0.0.0.0/0        0.0.0.0/0
2     0 0 ufw-before-forward all  -- *      *        0.0.0.0/0        0.0.0.0/0
3     0 0 ufw-after-forward all  -- *      *        0.0.0.0/0        0.0.0.0/0
4     0 0 ufw-after-logging-forward all  -- *      *        0.0.0.0/0        0.0.0.0/0
5     0 0 ufw-reject-forward all  -- *      *        0.0.0.0/0        0.0.0.0/0
6     0 0 ufw-track-forward all  -- *      *        0.0.0.0/0        0.0.0.0/0

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target     prot opt in     out     source            destination
1   4826 444K ufw-before-logging-output all  -- *      *        0.0.0.0/0        0.0.0.0/0
2   4826 444K ufw-before-output all  -- *      *        0.0.0.0/0        0.0.0.0/0
3     4 1224 ufw-after-output all  -- *      *        0.0.0.0/0        0.0.0.0/0
4     4 1224 ufw-after-logging-output all  -- *      *        0.0.0.0/0        0.0.0.0/0
5     4 1224 ufw-reject-output all  -- *      *        0.0.0.0/0        0.0.0.0/0
6     4 1224 ufw-track-output all  -- *      *        0.0.0.0/0        0.0.0.0/0

Chain ufw-after-forward (1 references)

```

```

santi@ubutu
Session Actions Edit View Help
num  pkts bytes target    prot opt in     out    source      destination
Chain ufw-after-input (1 references)
num  pkts bytes target    prot opt in     out    source      destination
1    0     0 ufw-skip-to-policy-input  udp  --  *      *          0.0.0.0/0    0.0.0.0/0
    udp dpt:137
2    0     0 ufw-skip-to-policy-input  udp  --  *      *          0.0.0.0/0    0.0.0.0/0
    udp dpt:138
3    0     0 ufw-skip-to-policy-input  tcp  --  *      *          0.0.0.0/0    0.0.0.0/0
    tcp dpt:139
4    0     0 ufw-skip-to-policy-input  tcp  --  *      *          0.0.0.0/0    0.0.0.0/0
    tcp dpt:445
5    1    316 ufw-skip-to-policy-input  udp  --  *      *          0.0.0.0/0    0.0.0.0/0
    udp dpt:67
6    0     0 ufw-skip-to-policy-input  udp  --  *      *          0.0.0.0/0    0.0.0.0/0
    udp dpt:68
7    43   3096 ufw-skip-to-policy-input  all  --  *      *          0.0.0.0/0    0.0.0.0/0
    ADDRTYPE match dst-type BROADCAST

Chain ufw-after-logging-forward (1 references)
num  pkts bytes target    prot opt in     out    source      destination      limit: av
1    0     0 LOG      all  --  *      *          0.0.0.0/0    0.0.0.0/0
g 3/min burst 10 LOG flags 0 level 4 prefix "[UFW BLOCK] "

Chain ufw-after-logging-input (1 references)
num  pkts bytes target    prot opt in     out    source      destination      limit: av
1    3    180 LOG      all  --  *      *          0.0.0.0/0    0.0.0.0/0
g 3/min burst 10 LOG flags 0 level 4 prefix "[UFW BLOCK] "

Chain ufw-after-logging-output (1 references)
num  pkts bytes target    prot opt in     out    source      destination

Chain ufw-after-output (1 references)
num  pkts bytes target    prot opt in     out    source      destination

Chain ufw-before-forward (1 references)
num  pkts bytes target    prot opt in     out    source      destination      ctstate R
1    0     0 ACCEPT   all  --  *      *          0.0.0.0/0    0.0.0.0/0
RELATED,ESTABLISHED
2    0     0 ACCEPT   icmp --  *      *          0.0.0.0/0    0.0.0.0/0      icmp type
3
3    0     0 ACCEPT   icmp --  *      *          0.0.0.0/0    0.0.0.0/0      icmp type
11
```

```

Session Actions Edit View Help
Chain ufw-before-input (1 references)
num  pkts bytes target    prot opt in     out     source           destination
1      648 48240 ACCEPT    all  --  lo      *      0.0.0.0/0        0.0.0.0/0
2      4203 375K ACCEPT    all  --  *       *      0.0.0.0/0        0.0.0.0/0          ctstate R
RELATED,ESTABLISHED
3      0      0 ufw-logging-deny all  --  *       *      0.0.0.0/0        0.0.0.0/0          ct
state INVALID
4      0      0 DROP      all  --  *       *      0.0.0.0/0        0.0.0.0/0          ctstate I
INVALID
5      0      0 ACCEPT    icmp --  *       *      0.0.0.0/0        0.0.0.0/0          icmptype
3
6      0      0 ACCEPT    icmp --  *       *      0.0.0.0/0        0.0.0.0/0          icmptype
11
7      0      0 ACCEPT    icmp --  *       *      0.0.0.0/0        0.0.0.0/0          icmptype
12
8      0      0 ACCEPT    icmp --  *       *      0.0.0.0/0        0.0.0.0/0          icmptype
8
9      1      576 ACCEPT    udp  --  *       *      0.0.0.0/0        0.0.0.0/0          udp spt:6
7 dpt:68
10     54    4012 ufw-not-local all  --  *       *      0.0.0.0/0        0.0.0.0/0
11     0      0 ACCEPT    udp  --  *       *      0.0.0.0/0        224.0.0.251        udp dpt:5
353
12     0      0 ACCEPT    udp  --  *       *      0.0.0.0/0        239.255.255.250    udp dpt:1
900
13     54    4012 ufw-user-input all  --  *       *      0.0.0.0/0        0.0.0.0/0

Chain ufw-before-logging-forward (1 references)
num  pkts bytes target    prot opt in     out     source           destination

Chain ufw-before-logging-input (1 references)
num  pkts bytes target    prot opt in     out     source           destination

Chain ufw-before-logging-output (1 references)
num  pkts bytes target    prot opt in     out     source           destination

Chain ufw-before-output (1 references)
num  pkts bytes target    prot opt in     out     source           destination
1      648 48240 ACCEPT    all  --  *       lo      0.0.0.0/0        0.0.0.0/0
2      4174 394K ACCEPT    all  --  *       *      0.0.0.0/0        0.0.0.0/0          ctstate R
RELATED,ESTABLISHED
3      4      1224 ufw-user-output all  --  *       *      0.0.0.0/0        0.0.0.0/0

Chain ufw-logging-allow (0 references)

```

```
1 2 3 4 5
santi@ubu

Session Actions Edit View Help

num pkts bytes target prot opt in out source destination
Chain ufw-track-input (1 references)
num pkts bytes target prot opt in out source destination
Chain ufw-track-output (1 references)
num pkts bytes target prot opt in out source destination
1 0 0 ACCEPT tcp -- * * 0.0.0.0/0 0.0.0.0/0 ctstate N
EW
2 4 1224 ACCEPT udp -- * * 0.0.0.0/0 0.0.0.0/0 ctstate N
EW
Chain ufw-user-forward (1 references)
num pkts bytes target prot opt in out source destination
Chain ufw-user-input (1 references)
num pkts bytes target prot opt in out source destination
1 0 0 ACCEPT tcp -- * * 0.0.0.0/0 0.0.0.0/0 tcp dpt:22
2
Chain ufw-user-limit (0 references)
num pkts bytes target prot opt in out source destination
1 0 0 LOG all -- * * 0.0.0.0/0 0.0.0.0/0 limit: avg
g 3/min burst 5 LOG flags 0 level 4 prefix "[UFW LIMIT BLOCK] "
2 0 0 REJECT all -- * * 0.0.0.0/0 0.0.0.0/0 reject-wi
th icmp-port-unreachable
Chain ufw-user-limit-accept (0 references)
num pkts bytes target prot opt in out source destination
1 0 0 ACCEPT all -- * * 0.0.0.0/0 0.0.0.0/0
Chain ufw-user-logging-forward (0 references)
num pkts bytes target prot opt in out source destination
Chain ufw-user-logging-input (0 references)
num pkts bytes target prot opt in out source destination
Chain ufw-user-logging-output (0 references)
num pkts bytes target prot opt in out source destination
Chain ufw-user-output (1 references)
num pkts bytes target prot opt in out source destination
santi@ubutu:~$
```

```

Session Actions Edit View Help
santi@ubuntu:~$ sudo iptables -D INPUT -p tcp --dport 9999 -j LAB4_TEST
sudo iptables -F LAB4_TEST
sudo iptables -X LAB4_TEST
sudo iptables -L -n -v --line-numbers | tee ~/core102_lab4/12_iptables_after_cleanup.txt
sudo ufw status numbered
Chain INPUT (policy DROP 9 packets, 540 bytes)
num  pkts bytes target     prot opt in     out     source            destination
1    5956 519K ufw-before-logging-input all  --  *      *        0.0.0.0/0         0.0.0.0/0
2    5956 519K ufw-before-input all  --  *      *        0.0.0.0/0         0.0.0.0/0
3     93 6832 ufw-after-input all  --  *      *        0.0.0.0/0         0.0.0.0/0
4     9  540 ufw-after-logging-input all  --  *      *        0.0.0.0/0         0.0.0.0/0
5     9  540 ufw-reject-input all  --  *      *        0.0.0.0/0         0.0.0.0/0
6     9  540 ufw-track-input all  --  *      *        0.0.0.0/0         0.0.0.0/0

Chain FORWARD (policy DROP 0 packets, 0 bytes)
num  pkts bytes target     prot opt in     out     source            destination
1     0    0 ufw-before-logging-forward all  --  *      *        0.0.0.0/0         0.0.0.0/0
2     0    0 ufw-before-forward all  --  *      *        0.0.0.0/0         0.0.0.0/0
3     0    0 ufw-after-forward all  --  *      *        0.0.0.0/0         0.0.0.0/0
4     0    0 ufw-after-logging-forward all  --  *      *        0.0.0.0/0         0.0.0.0/0
5     0    0 ufw-reject-forward all  --  *      *        0.0.0.0/0         0.0.0.0/0
6     0    0 ufw-track-forward all  --  *      *        0.0.0.0/0         0.0.0.0/0

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target     prot opt in     out     source            destination
1   5826 551K ufw-before-logging-output all  --  *      *        0.0.0.0/0         0.0.0.0/0
2   5826 551K ufw-before-output all  --  *      *        0.0.0.0/0         0.0.0.0/0
3     8  2448 ufw-after-output all  --  *      *        0.0.0.0/0         0.0.0.0/0
4     8  2448 ufw-after-logging-output all  --  *      *        0.0.0.0/0         0.0.0.0/0
5     8  2448 ufw-reject-output all  --  *      *        0.0.0.0/0         0.0.0.0/0
6     8  2448 ufw-track-output all  --  *      *        0.0.0.0/0         0.0.0.0/0

Chain ufw-after-forward (1 references)
num  pkts bytes target     prot opt in     out     source            destination

Chain ufw-after-input (1 references)
num  pkts bytes target     prot opt in     out     source            destination

```

```

Session Actions Edit View Help
g 3/min burst 10 LOG flags 0 level 4 prefix "[UFW BLOCK] "

Chain ufw-after-logging-output (1 references)
num  pkts bytes target    prot opt in     out     source               destination

Chain ufw-after-output (1 references)
num  pkts bytes target    prot opt in     out     source               destination

Chain ufw-before-forward (1 references)
num  pkts bytes target    prot opt in     out     source               destination
1      0      0 ACCEPT    all  --  *      *      0.0.0.0/0           0.0.0.0/0           ctstate R
RELATED,ESTABLISHED
2      0      0 ACCEPT    icmp --  *      *      0.0.0.0/0           0.0.0.0/0           icmptype
3
3      0      0 ACCEPT    icmp --  *      *      0.0.0.0/0           0.0.0.0/0           icmptype
11
4      0      0 ACCEPT    icmp --  *      *      0.0.0.0/0           0.0.0.0/0           icmptype
12
5      0      0 ACCEPT    icmp --  *      *      0.0.0.0/0           0.0.0.0/0           icmptype
8
6      0      0 ufw-user-forward all  --  *      *      0.0.0.0/0           0.0.0.0/0

Chain ufw-before-input (1 references)
num  pkts bytes target    prot opt in     out     source               destination
1      728 54208 ACCEPT    all  --  lo     *      0.0.0.0/0           0.0.0.0/0
2      5130 458K ACCEPT    all  --  *      *      0.0.0.0/0           0.0.0.0/0           ctstate R
RELATED,ESTABLISHED
3      0      0 ufw-logging-deny all  --  *      *      0.0.0.0/0           0.0.0.0/0           ct
state INVALID
4      0      0 DROP      all  --  *      *      0.0.0.0/0           0.0.0.0/0           ctstate I
INVALID
5      0      0 ACCEPT    icmp --  *      *      0.0.0.0/0           0.0.0.0/0           icmptype
3
6      0      0 ACCEPT    icmp --  *      *      0.0.0.0/0           0.0.0.0/0           icmptype
11
7      0      0 ACCEPT    icmp --  *      *      0.0.0.0/0           0.0.0.0/0           icmptype
12
8      0      0 ACCEPT    icmp --  *      *      0.0.0.0/0           0.0.0.0/0           icmptype
8
9      1      576 ACCEPT    udp   --  *      *      0.0.0.0/0           0.0.0.0/0           udp spt:6
7 dpt:68
10     97 7072 ufw-not-local all  --  *      *      0.0.0.0/0           0.0.0.0/0
11     0      0 ACCEPT    udp   --  *      *      0.0.0.0/0           224.0.0.251         udp dpt:5

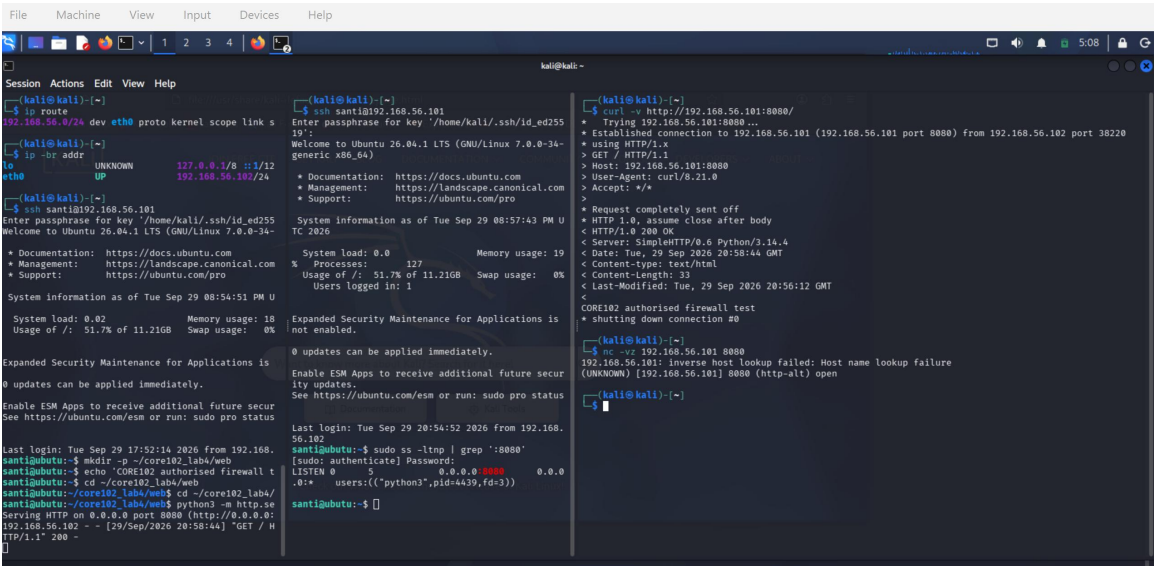
```

```

Session Actions Edit View Help
Chain ufw-reject-input (1 references)
num  pkts bytes target    prot opt in     out     source           destination
Chain ufw-reject-output (1 references)
num  pkts bytes target    prot opt in     out     source           destination
Chain ufw-skip-to-policy-forward (0 references)
num  pkts bytes target    prot opt in     out     source           destination
1      0      0 DROP      all  --  *      *      0.0.0.0/0        0.0.0.0/0
Chain ufw-skip-to-policy-input (7 references)
num  pkts bytes target    prot opt in     out     source           destination
1     84 6292 DROP      all  --  *      *      0.0.0.0/0        0.0.0.0/0
Chain ufw-skip-to-policy-output (0 references)
num  pkts bytes target    prot opt in     out     source           destination
1      0      0 ACCEPT    all  --  *      *      0.0.0.0/0        0.0.0.0/0
Chain ufw-track-forward (1 references)
num  pkts bytes target    prot opt in     out     source           destination
Chain ufw-track-input (1 references)
num  pkts bytes target    prot opt in     out     source           destination
Chain ufw-track-output (1 references)
num  pkts bytes target    prot opt in     out     source           destination   ctstate
1      0      0 ACCEPT    tcp  --  *      *      0.0.0.0/0        0.0.0.0/0                  N
EW
2      8 2448 ACCEPT    udp  --  *      *      0.0.0.0/0        0.0.0.0/0                  N
EW
Chain ufw-user-forward (1 references)
num  pkts bytes target    prot opt in     out     source           destination
Chain ufw-user-input (1 references)
num  pkts bytes target    prot opt in     out     source           destination   tcp dpt:2
1      0      0 ACCEPT    tcp  --  *      *      0.0.0.0/0        0.0.0.0/0
2
Chain ufw-user-limit (0 references)
num  pkts bytes target    prot opt in     out     source           destination   limit: av
1      0      0 LOG      all  --  *      *      0.0.0.0/0        0.0.0.0/0
g 3/min burst 5 LOG flags 0 level 4 prefix "[UFW LIMIT BLOCK] "
```

Part 2 Controlled Test Service

Evidence



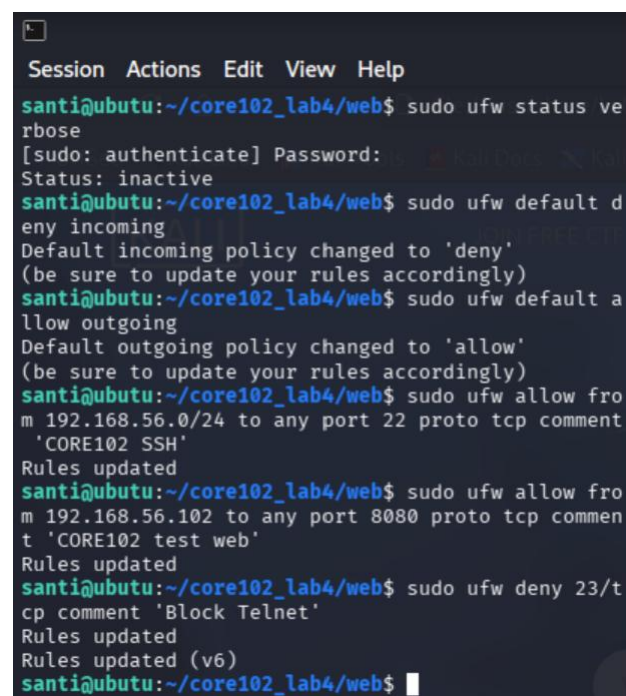
OBJ:OBJ

Report

I created a temporary test directory and page on the server and started a Python HTTP server bound to 0.0.0.0 on TCP port 8080. I confirmed the listener with `ss -ltnp`, which showed `python3` bound to `0.0.0.0:8080`. From the Kali client at `192.168.56.102` I tested the connection before any firewall changes using both `curl -v` and `nc`. Both succeeded: the HTTP response was 200 OK with the page contents CORE102 authorised firewall test, and `nc` reported port 8080 open. The server log recorded the GET request from `192.168.56.102`, confirming the client could reach the server at baseline before UFW was applied.

Part 3 UFW Policy Implementation

Evidence

A terminal window with a dark background and light-colored text. The prompt is `santi@ubuntu:~/core102_lab4/web$`. The user enters `sudo ufw status verbose`, followed by `sudo ufw default deny incoming`, `sudo ufw default allow outgoing`, `sudo ufw allow from 192.168.56.0/24 to any port 22 proto tcp comment 'CORE102 SSH'`, `sudo ufw allow from 192.168.56.102 to any port 8080 proto tcp comment 'CORE102 test web'`, and `sudo ufw deny 23/tcp comment 'Block Telnet'`. Each command is followed by a confirmation message from UFW. The terminal ends with the prompt `santi@ubuntu:~/core102_lab4/web$` and a cursor.

```
Session  Actions  Edit  View  Help
santi@ubuntu:~/core102_lab4/web$ sudo ufw status ve
rbose
[sudo: authenticate] Password:
Status: inactive
santi@ubuntu:~/core102_lab4/web$ sudo ufw default d
eny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
santi@ubuntu:~/core102_lab4/web$ sudo ufw default a
llow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
santi@ubuntu:~/core102_lab4/web$ sudo ufw allow fro
m 192.168.56.0/24 to any port 22 proto tcp comment
'CORE102 SSH'
Rules updated
santi@ubuntu:~/core102_lab4/web$ sudo ufw allow fro
m 192.168.56.102 to any port 8080 proto tcp commen
t 'CORE102 test web'
Rules updated
santi@ubuntu:~/core102_lab4/web$ sudo ufw deny 23/t
cp comment 'Block Telnet'
Rules updated
Rules updated (v6)
santi@ubuntu:~/core102_lab4/web$
```

```
santi@ubutu:~/core102_lab4/web$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? yes
Firewall is active and enabled on system startup
santi@ubutu:~/core102_lab4/web$
```

```
santi@ubutu:~/core102_lab4/web$ sudo ufw status numbered
Status: active

    To Action From
--
[ 1] 22/tcp ALLOW IN Anywhere
    # Allow SSH
[ 2] 192.168.56.102 8080 DENY OUT Anywhere
    (out)
[ 3] 22/tcp ALLOW IN 192.168.56.0/24
    # CORE102 SSH
[ 4] 8080/tcp ALLOW IN 192.168.56.102
    # CORE102 test web
[ 5] 23/tcp DENY IN Anywhere
    # Block Telnet
[ 6] 22/tcp (v6) ALLOW IN Anywhere
    # Allow SSH
[ 7] 23/tcp (v6) DENY IN Anywhere
    # Block Telnet

santi@ubutu:~/core102_lab4/web$ sudo ss -ltnp
State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
LISTEN 0 4096 0.0.0.0:22 0.0.0.0:* users:(("sshd",pid=1202,fd=3),("systemd",pid=1,fd=224))
LISTEN 0 4096 127.0.0.54:53 0.0.0.0:* users:(("systemd-resolve",pid=584,fd=19))
LISTEN 0 4096 127.0.0.53%lo:53 0.0.0.0:* users:(("systemd-resolve",pid=584,fd=17))
LISTEN 0 4096 [::]:22 [::]:* users:(("sshd",pid=1202,fd=4),("systemd",pid=1,fd=225))
santi@ubutu:~/core102_lab4/web$
```

```

Session  Actions  Edit  View  Help
State    Recv-Q    Send-Q    Local Address:Port
Peer Address:Port
Process
LISTEN    0          4096      0.0.0.0:22
          0.0.0.0:*
          users:((("sshd",pid=1202,fd=3),("systemd",pid=1,fd=224))
LISTEN    0          4096      127.0.0.54:53
          0.0.0.0:*
          users:((("systemd-resolve",pid=584,fd=19))
LISTEN    0          4096      127.0.0.53%lo:53
          0.0.0.0:*
          users:((("systemd-resolve",pid=584,fd=17))
LISTEN    0          4096      [::]:22
          [::]:*
          users:((("sshd",pid=1202,fd=4),("systemd",pid=1,fd=225))
santi@ubutu:~/core102_lab4/web$ sudo ufw delete 2
Deleting:
deny out to 192.168.56.102 port 8080
Proceed with operation (y|n)? yes
Rule deleted
santi@ubutu:~/core102_lab4/web$ sudo ufw status numbered
Status: active

    To Action From
--
[ 1] 22/tcp      ALLOW IN Anywhere
    # Allow SSH
[ 2] 22/tcp      ALLOW IN 192.168.56.0/24
    # CORE102 SSH
[ 3] 8080/tcp     ALLOW IN 192.168.56.102
    # CORE102 test web
[ 4] 23/tcp      DENY IN Anywhere
    # Block Telnet
[ 5] 22/tcp (v6) ALLOW IN Anywhere
    # Allow SSH
[ 6] 23/tcp (v6) DENY IN Anywhere
    # Block Telnet
santi@ubutu:~/core102_lab4/web$

```

Report

I confirmed UFW was inactive with `ufw status verbose` before making changes. I set the default incoming policy to deny and the default outgoing policy to allow, then added an allow rule for SSH from the private lab subnet (192.168.56.0/24) on port 22, a narrower allow for the temporary web service on 8080 restricted to the authorised client (192.168.56.102), and an explicit deny for Telnet on port 23. I enabled UFW and confirmed with yes at the prompt. `sudo ufw status numbered` showed SSH allowed from the subnet, port 8080 allowed only from 192.168.56.102, and Telnet denied on both IPv4 and IPv6. I noticed an unintended DENY OUT rule for 8080 had been created during typing, deleted it by its numbered index, and re-listed to confirm the clean allow-only policy. `ss -ltnp` confirmed `sshd` was still listening on 0.0.0.0:22 and [::]:22, showing UFW filters traffic without closing sockets.

Part 4 Rule Verification

Test	Expected	Observed	Matching rule or explanation
SSH from client	Allowed	open	Rule 2: ALLOW IN 22 from 192.168.56.0/24
HTTP 8080 from authorised client	Allowed	open + curl → CORE102 authorised firewall test	Rule 3: ALLOW IN 8080 from 192.168.56.102
HTTP 8080 from other client	Denied	Not tested (no 3rd VM) — default deny drops	Default deny incoming
Telnet 23	Denied	time out	Rule 4: DENY IN 23/tcp + default deny

Evidence

Report

From the authorised client at 192.168.56.102 I tested four things against the server. SSH on port 22 returned open, matching UFW rule 2 which allows port 22 from 192.168.56.0/24. HTTP on port 8080 initially returned connection refused because the Python test server was not running; after restarting it with `python3 -m http.server 8080 --bind 0.0.0.0` and confirming with `ss -ltnp`, port 8080 returned open from the client and curl returned HTTP 200 with the page contents CORE102 authorised firewall test, matching rule 3 which allows 8080 only from 192.168.56.102. A TCP probe on port 23 (Telnet) timed out, matching rule 4 which

explicitly denies 23/tcp combined with the default-deny incoming policy. The server-side UFW packet counters showed traffic being processed through the ufw-before-input, ufw-user-input, ufw-after-input, ufw-reject-input, and ufw-track-input chains with policy DROP, and the server's HTTP log recorded the GET request from 192.168.56.102, confirming the authorised flow was permitted end-to-end.

Part 5 Safe iptables Chain Exercise

Evidence

```
santi@ubuntu:~/core102_lab4/web$ sudo iptables -N CORE102_LAB
santi@ubuntu:~/core102_lab4/web$ sudo iptables -A CORE102_LAB -p tcp --dport 23 -j DROP
santi@ubuntu:~/core102_lab4/web$ sudo iptables -A CORE102_LAB -p tcp --dport 22 -j ACCEPT
santi@ubuntu:~/core102_lab4/web$ sudo iptables -L CORE102_LAB -n -v --line-numbers
Chain CORE102_LAB (0 references)
num  pkts bytes target     prot opt in     out     source            destination        tcp dpt:2
1      0      0 DROP      tcp  --  *      *        0.0.0.0/0         0.0.0.0/0
2      0      0 ACCEPT    tcp  --  *      *        0.0.0.0/0         0.0.0.0/0
santi@ubuntu:~/core102_lab4/web$ sudo iptables -S CORE102_LAB
-N CORE102_LAB
-A CORE102_LAB -p tcp -m tcp --dport 23 -j DROP
-A CORE102_LAB -p tcp -m tcp --dport 22 -j ACCEPT
santi@ubuntu:~/core102_lab4/web$ sudo iptables -F CORE102_LAB
santi@ubuntu:~/core102_lab4/web$ sudo iptables -X CORE102_LAB
santi@ubuntu:~/core102_lab4/web$ sudo iptables -L CORE102_LAB 2>&1 || echo "Custom chain removed"
iptables: No chain/target/match by that name.
Custom chain removed
santi@ubuntu:~/core102_lab4/web$
```

Report

I created a user-defined iptables chain called CORE102_LAB and appended two rules to it: DROP for TCP destination port 23 (Telnet), then ACCEPT for TCP destination port 22 (SSH). `iptables -L CORE102_LAB -n -v --line-numbers` showed the chain with 0 references — meaning no built-in chain was jumping into it — and both rules listed in order. `iptables -S CORE102_LAB` confirmed the exact rule syntax: `-A CORE102_LAB -p tcp -m tcp --dport 23 -j DROP` and `-A CORE102_LAB -p tcp -m tcp --dport 22 -j ACCEPT`. I then flushed the chain with `iptables -F CORE102_LAB` and deleted it with `iptables -X CORE102_LAB`. The final verification command `iptables -L CORE102_LAB 2>&1 || echo "Custom chain removed"` printed `iptables: No chain/target/match by that name followed by Custom chain removed`, confirming the chain no longer exists.

Part 6 SSH Key Authentication

Evidence

```
(kali㉿kali)-[~]
$ mkdir -p ~/.ssh

(kali㉿kali)-[~]
$ chmod 700 ~/.ssh

(kali㉿kali)-[~]
$ ssh-keygen -t ed25519 -a 64 -f ~/.ssh/core102_ed25519 -C "CORE102 authorised lab"
Generating public/private ed25519 key pair.
Enter passphrase for "/home/kali/.ssh/core102_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/kali/.ssh/core102_ed25519
Your public key has been saved in /home/kali/.ssh/core102_ed25519.pub
The key fingerprint is:
SHA256:e4NcCgiOGzU41g+jRHxtrCn8oXZwC4sUDT301D/1SgE CORE102 authorised lab
The key's randomart image is:
+--[ED25519 256]--+
|.o=.o.. E. |
|.+.o+ . . o |
|+0*+=. . . o |
|o@.B+. o . . |
|=.X +.. S + . |
|. * + o = . |
|o . = o |
| . . |
+-----[SHA256]-----+

(kali㉿kali)-[~]
$
```

```
(kali㉿kali)-[~]
$ ssh-keygen -lf ~/.ssh/core102_ed25519.pub
256 SHA256:e4NcCgiOGzU41g+jRHxtrCn8oXZwC4sUDT301D/1SgE CORE102 authorised lab (ED25519)

(kali㉿kali)-[~]
$ ssh-copy-id -i ~/.ssh/core102_ed25519.pub santi@192.168.56.101
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/kali/.ssh/core102_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
Enter passphrase for key '/home/kali/.ssh/id_ed25519':

Number of key(s) added: 1

Now try logging into the machine, with: "ssh -i /home/kali/.ssh/core102_ed25519 'santi@192.168.56.101'"
and check to make sure that only the key(s) you wanted were added.

(kali㉿kali)-[~]
$ ssh -i ~/.ssh/core102_ed25519 santi@192.168.56.101
Enter passphrase for key '/home/kali/.ssh/core102_ed25519':
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-34-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Sep 29 10:01:33 PM UTC 2026

System load: 0.05           Memory usage: 18%    Processes:      132
Usage of /:  51.7% of 11.21GB Swap usage:   0%    Users logged in: 1

Expanded Security Maintenance for Applications is not enabled.
```

```
(kali㉿kali)-[~]
$ ssh-copy-id -i ~/.ssh/core102_ed25519.pub santi@192.168.56.101
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/kali/.ssh/core102_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
Enter passphrase for key '/home/kali/.ssh/id_ed25519':

Number of key(s) added: 1

Now try logging into the machine, with: "ssh -i /home/kali/.ssh/core102_ed25519 'santi@192.168.56.101'"
and check to make sure that only the key(s) you wanted were added.

(kali㉿kali)-[~]
$ ssh -i ~/.ssh/core102_ed25519 santi@192.168.56.101
Enter passphrase for key '/home/kali/.ssh/core102_ed25519':
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-34-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Sep 29 10:01:33 PM UTC 2026

System load: 0.05           Memory usage: 18%    Processes:      132
Usage of /:  51.7% of 11.21GB Swap usage:  0%      Users logged in: 1

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Sep 29 21:52:50 2026 from 192.168.56.102
santi@ubutu:~$
```

Report

On the Kali client I created a dedicated Ed25519 key pair with a strong passphrase, using -a 64 for additional KDF rounds and the comment "CORE102 authorised lab".

The public key fingerprint was recorded as

SHA256:e4NCcgi0GzU41g+jRHxtrCn80xZwc4sUDT301D/1SgE (ED25519) — the private key was never displayed or copied. I deployed the public key with ssh-copy-id, which reported "Number of key(s) added: 1". I then tested a new SSH session with `ssh -i ~/.ssh/core102_ed25519 santi@192.168.56.101` — the client was prompted for the key passphrase (not the account password) and logged in successfully, confirming the key pair is functional. On the server, `journalctl -u ssh` recorded Accepted publickey for santi from 192.168.56.102 with the matching ED25519 fingerprint, and `stat` confirmed permissions of 700 on `~/.ssh` and 600 on `~/.ssh/authorized_keys`.

Part 7 SSH Server Hardening

Evidence

kali (lab4- baseline) [Running] - Oracle VirtualBox

```
File Machine View Input Devices Help

santi@ubutu: ~/core102_lab4/web$ sudo cp /etc/ssh/sshd_config /etc/ssh/sshd_config.core102.bak
[sudo: authenticate] Password:
santi@ubutu:~/core102_lab4/web$ sudo nano /etc/ssh/sshd_config.d/90-core102-hardening.conf
santi@ubutu:~/core102_lab4/web$ sudo sshd -t
[sudo: authenticate] Password:
santi@ubutu:~/core102_lab4/web$ sudo systemctl reload ssh
santi@ubutu:~/core102_lab4/web$ sudo systemctl --no-pager status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-09-29 15:22:27 UTC; 7h ago
     Invocation: 4750e3626f044bd2979d14ac44cbaa72
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Process: 5883 ExecReload=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
    Process: 5886 ExecReload=/bin/kill -HUP $MAINPID (code=exited, status=0/SUCCESS)
   Main PID: 1202 (sshd)
      Tasks: 1 (limit: 1888)
    Memory: 10.9M (peak: 12.7M)
       CPU: 843ms
    CGroup: /system.slice/ssh.service
            └─1202 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 29 22:01:28 ubutu sshd-session[5551]: Connection closed by authenticating user santi 192.168...eauth]
Sep 29 22:01:33 ubutu sshd-session[5553]: Accepted publickey for santi from 192.168.56.102 port ..RGLSRc
Sep 29 22:01:33 ubutu sshd-session[5553]: pam_unix(sshd:session): session opened for user santi(....uid=0)
Sep 29 22:02:05 ubutu sshd-session[5661]: Accepted publickey for santi from 192.168.56.102 port ...D/1SgE
Sep 29 22:02:05 ubutu sshd-session[5661]: pam_unix(sshd:session): session opened for user santi(....uid=0)
Sep 29 23:01:11 ubutu systemd[1]: Reloading ssh.service - OpenBSD Secure Shell server...
Sep 29 23:01:11 ubutu sshd[1202]: Received SIGHUP; restarting.
Sep 29 23:01:11 ubutu systemd[1]: Reloaded ssh.service - OpenBSD Secure Shell server.
Sep 29 23:01:11 ubutu sshd[1202]: Server listening on 0.0.0.0 port 22.
Sep 29 23:01:11 ubutu sshd[1202]: Server listening on :: port 22.
Hint: Some lines were ellipsized, use -l to show in full.
santi@ubutu:~/core102_lab4/web$ cat /etc/ssh/sshd_config.d/90-core102-hardening.conf
PermitRootLogin no
PubkeyAuthentication yes
PasswordAuthentication no
KbdInteractiveAuthentication no

santi@ubutu:~/core102_lab4/web$
```

Report

I backed up the SSH server config to `/etc/ssh/sshd_config.core102.bak`, then created a drop-in file at `/etc/ssh/sshd_config.d/90-core102-hardening.conf` containing four hardening settings: `PermitRootLogin no`, `PubkeyAuthentication yes`, `PasswordAuthentication no`, and `KbdInteractiveAuthentication no`. I validated the config with `sudo sshd -t`, which returned no output (meaning valid), and only then reloaded SSH with `sudo systemctl reload ssh`. `systemctl status ssh` confirmed the service was active and had been reloaded cleanly, and the journal recorded Server listening on 0.0.0.0 port 22 after the reload. With my original session still open, I opened a new Kali terminal and logged in with `ssh -i ~/.ssh/core102_ed25519 santi@192.168.56.101` — the client was prompted for the key passphrase only and the session opened successfully, proving key-based authentication still works with password auth disabled. The cat output of the drop-in file confirms the four settings are in effect.

Part 8 Final Exposure Review and Cleanup

Evidence

confirmed the socket was gone with `sudo ss -ltnp | grep ':8080'`, which printed "Temporary web service stopped". I listed the UFW rules with `sudo ufw status numbered`, deleted the temporary port 8080 allow rule by its index (rule 3), and re-listed immediately to confirm removal — the remaining rules were only the SSH allow rules (port 22 from 192.168.56.0/24 on IPv4 and IPv6) and the Telnet deny rules. The final `ss -tulnp` inventory showed exactly the same listeners as the baseline: `sshd` on 0.0.0.0:22 and `:::22`, `systemd-resolve` on 127.0.0.53/54 (loopback only), `systemd-networkd` on port 68 (DHCP client reply socket), and `chronyd` on 127.0.0.1/[`::1`]:323 (loopback only). Port 8080 no longer appears. Every remaining listener matches its baseline purpose: SSH is the only network-reachable service, and it is now restricted to the lab subnet by UFW. The client private key remains at `~/.ssh/core102_ed25519` pending instructor confirmation of evidence review.

Reflection Questions

1. Why must SSH access be allowed before a default-deny firewall is enabled?

Once default-deny incoming is active, all new inbound connections are dropped unless an allow rule matches first. If the SSH allow isn't already in place, a reconnect after the current session drops would be blocked, leaving only the VM console for recovery. Allow SSH first, then enable.

2. Difference between an open port and an authorised service?

An open port is any socket in LISTEN state. An authorised service is one deliberately permitted by policy — documented purpose plus a matching firewall allow rule. A port can be open yet blocked, and a service can be authorised but stopped.

3. Why does the server need the public key while the client must protect the private key?

The server stores the client's public key to verify signatures during authentication. Only the client's private key can produce those signatures — if it leaks, anyone can impersonate the client. The public key alone cannot authenticate.

4. Which rollback step recovers access after an invalid SSH configuration?

From the VM console: remove `/etc/ssh/sshd_config.d/90-core102-hardening.conf`, restore `/etc/ssh/sshd_config.core102.bak` over the main config, run `sudo sshd -t` to

validate, then `sudo systemctl reload ssh`.

Final Checklist

Worked in authorised lab only

Add full name + registration number to report

Commands and results visible in evidence

Results explained, not just screenshots

Strip passwords/private keys/personal info before submitting

Open PDF + ZIP to confirm completeness